Laser Ablation Inductively Coupled Plasma Mass Spectrometry; Grain exterior (spot size 25 µm)

Sample	Core	Al (mg/kg)	P (mg/kg)	Sr (mg/kg)	Y (mg/kg)	Ba (mg/kg)	La (mg/kg)	Ce (mg/kg)	Pr (mg/kg)	Nd (mg/kg)	Sm (mg/kg)	Eu (mg/kg)	Gd (mg/kg)	Tb (mg/kg)	Dy (mg/kg)	Ho (mg/kg)	Er (mg/kg)	Tm (mg/kg)	Yb (mg/kg)	Lu (mg/kg)	Th (mg/kg)	U238 (mg/kg)
SM1	25005	4552.63	183277.16	3411.94	537.16	443.72	212.85	376.99	63.38	209.38	43.09	8.27	76.39	7.88	49.03	10.09	36.19	4.72	37.98	5.85	23.81	114.72
SM2	25005	6767.20	132467.44	3750.19	283.49	517.16	109.84	184.29	23.89	87.68	22.05	5.30	24.74	3.65	28.32	6.32	20.93	3.05	21.80	3.48	11.06	71.98
SM3	25005	2541.83	169056.56	3796.97	84.70	705.39	30.20	38.45	4.32	17.06	2.24	0.76	2.88	0.48	5.09	1.42	5.57	0.78	7.32	1.16	1.22	44.17
SM4	25005	4757.70	150583.77	3731.24	336.74	183.03	127.88	221.19	31.98	135.49	29.62	6.92	36.78	5.00	35.44	7.92	25.24	3.72	23.47	4.00	15.59	113.64
SM5	25005	4502.02	166697.92	3788.70	13.23	200.73	5.07	8.93	1.04	3.91	1.35	<0.36	<2.19	0.15	1.14	0.30	0.58	0.23	1.03	<0.110	1	38.08
SM6	25005	5287.65	136853.44	3729.60	235.90	207.21	84.32	153.78	22.35	87.16	22.90	4.78	26.03	3.92	26.72	5.72	19.35	2.51	18.20	2.55	16.12	152.53
SM7	25005	1731.99	172774.05	3744.43	35.92	183.83	11.14	12.25	1.38	5.82	0.85	<0.30	1.42	0.28	1.63	0.64	2.08	0.38	2.41	0.53	0.53	54.31
SM8	25005	5617.72	146809.48	3657.16	304.82	139.37	107.13	185.70	26.85	112.83	24.28	5.71	30.06	4.43	32.52	7.59	23.03	3.26	23.62	3.74	11.5	80.93
SM9	25005	3910.31	175776.05	3833.81	29.91	134.84	10.68	10.41	1.13	4.45	0.56	0.24	2.41	0.16	1.15	0.44	1.56	0.30	2.93	0.38	0.87	50.76
SM10	25005	7662.71	136001.19	3775.75	484.98	184.76	177.96	318.88	47.33	211.07	44.47	11.18	60.66	8.67	56.76	11.87	37.32	5.93	37.62	5.34	37.66	159.14
SM11	25005	5615.60	180317.42	3926.67	23.06	291.96	6.22	8.85	0.98	4.63	0.83	0.41	<1.38	0.14	1.26	0.31	1.49	0.21	1.61	0.44	0.75	33.6
SM12	25005	4295.10	144408.42	4014.57	143.21	374.71	52.61	87.31	12.15	54.65	12.05	3.13	14.75	2.04	14.47	3.26	11.05	1.33	10.09	1.96	10.59	76.06
SM13	25005	5001.27	177022.41	4311.06	13.99	452.91	3.26	4.78	0.61	3.02	0.82	<0.35	<2.11	0.09	0.96	0.22	1.00	0.10	0.66	0.16	0.72	28.7
SM14	25005	5350.51	138500.91	3854.84	382.37	192.80	138.37	235.95	35.30	147.00	29.21	9.75	44.94	5.24	41.68	9.04	32.14	3.56	26.12	4.77	27.23	160.02
SM15	25005	7148.86	162056.14	4004.73	29.40	226.12	8.22	8.79	1.06	4.09	1.08	0.46	<2.00	0.07	1.81	0.62	1.69	0.24	1.92	0.34	0.96	36.22
SM16	25005	12833.25	137615.14	3768.65	693.83	240.10	281.60	484.20	72.49	283.13	70.79	19.66	76.75	11.03	73.66	15.76	49.64	6.82	50.40	8.00	58.71	202.85
SM17	25005	7547.96	169863.11	4087.81	49.64	150.17	15.14	16.88	2.35	7.95	1.95	0.75	<2.05	0.43	3.02	0.98	3.12	0.55	3.64	0.60	0.99	33.74
SM18	25005	4409.82	161425.64	3862.70	213.40	227.39	77.07	135.51	20.42	75.72	18.63	4.15	22.33	3.23	22.50	5.18	17.64	2.63	17.19	3.00	14.32	83.87
SM19	25005	3853.48	178475.83	4048.79	16.35	211.72	5.34	7.20	0.83	3.49	<0.44	<0.33	<1.68	0.18	0.93	0.24	0.80	0.17	1.23	0.34	0.56	35.15
SM20	25005	5291.78	168174.33	4162.48	152.44	257.61	53.63	91.41	12.80	58.19	11.61	2.58	13.58	2.07	14.54	3.41	12.12	1.66	12.49	1.99	8.46	80.68
SM21	25005	6836.00	171588.58	4464.42	13.79	296.20	5.56	10.98	1.29	5.12	0.95	0.39	1.73	0.11	0.85	0.23	0.74	0.15	0.75	0.15	1.02	41.58
SM22	25005	2814.27	138621.72	3527.90	293.82	209.56	129.79	232.23	30.75	127.97	26.45	8.22	31.73	5.14	33.59	7.25	23.49	3.73	21.88	3.77	31.49	124.93
SM23	25005	814.13	164634.13	3679.96	76.20	192.56	26.83	26.09	3.00	14.23	2.06	0.67	2.97	0.57	4.69	1.34	4.54	0.56	4.70	0.98	0.94	41.92
SM24	25005	11388.87	135613.31	3967.05	288.41	234.46	113.01	209.10	30.84	114.88	26.53	6.42	27.92	3.92	28.56	6.21	22.14	3.12	21.46	2.98	12.83	80.85
SM25	25005	4312.05	184900.38	3921.30	75.66	185.36	28.21	24.03	2.94	12.86	1.48	0.37	<1.68	0.46	3.02	1.14	4.51	0.71	5.03	0.91	0.73	47.7
SM26	25005	2633.30	140504.91	3845.22	257.69	194.73	92.66	159.97	22.30	96.63	19.78	5.73	26.91	4.03	25.77	5.53	18.47	2.88	21.19	3.20	15.81	109.14
SM27	25005	4107.20	167785.33	4006.01	23.16	207.85	5.91	9.47	1.01	4.00	1.10	0.50	1.90	0.17	1.12	0.53	1.16	0.28	1.33	0.39	1.02	28.35
SM28	25005	4227.57	161225.28	3884.43	287.15	245.96	98.72	160.83	22.81	96.57	21.39	5.35	25.78	3.92	26.86	6.31	22.47	2.61	22.16	3.39	11.73	114.3
SM29	25005	2929.60	173929.75	4036.37	35.10	278.69	11.26	10.17	1.12	5.54	0.85	<0.25	<1.53	0.14	1.88	0.45	2.47	0.30	2.97	0.52	0.34	59.11
SM30	25005	3725.09	146354.72	3618.86	473.59	162.54	167.77	283.42	39.32	167.96	34.71	9.55	45.51	6.60	48.81	10.86	33.72	4.99	36.59	5.40	15.77	150.8
SM31	25005	6174.11	162695.00	3713.72	76.19	252.20	26.75	30.43	3.65	15.52	2.48	0.61	<2.30	0.71	4.23	1.24	4.97	0.75	5.61	0.95	1.62	59.81
SM32	25005	4643.54	143160.31	3768.12	188.81	232.20	67.51	127.29	17.64	67.54	16.89	4.27	20.12	2.98	20.93	4.95	16.66	1.92	14.87	2.58	13.64	111.11
SM33	25005	6606.58	169588.73	3979.43	10.67	208.09	2.66	4.25	0.44	1.73	0.70	0.32	<1.50	0.14	0.70	0.19	0.91	0.09	1.02	0.12	0.91	23.77
SM34	25005	12046.01	130279.53	3471.86	216.17	232.35	83.58	146.47	20.81	87.71	20.11	3.94	20.72	3.34	21.29	5.39	16.26	2.15	15.45	2.81	17.2	119.44
SM35	25005	5385.98	157869.03	3921.92	12.92	255.98	4.45	5.35	0.80	3.70	0.66	<0.39	<1.96	0.12	1.12	0.29	0.51	0.13	1.31	0.37	0.79	28.55
SM36	25005	7176.05	149203.58	3692.18	526.65	191.90	190.23	334.37	48.17	208.45	44.10	12.45	55.14	8.57	56.36	13.71	41.01	5.86	42.11	6.24	30.73	229.27
SM37	25005	9849.65	143603.06	3660.61	512.94	189.25	180.63	320.59	49.11	214.48	46.08	10.80	55.56	8.55	56.74	12.66	39.33	5.78	36.14	5.98	40.08	171.84
SM38	25005	13222.86	154534.39	3844.88	396.17	213.34	134.93	230.27	31.49	134.33	28.90	7.18	37.28	5.50	40.40	9.20	28.88	4.16	28.40	4.90	17.7	81.73
SM39	25005	3886.95	164368.47	3873.44	581.71	261.87	208.60	376.74	49.39	213.19	45.86	11.78	60.04	8.54	63.13	13.74	46.17	6.05	45.10	7.61	32.64	147.32
SM40	25005	9381.65	142408.98	3129.90	677.27	340.10	368.83	853.76	99.06	423.77	99.95	21.23	107.38	13.70	93.44	17.99	53.99	7.38	49.48	8.20	225.42	322.03
SM41	CG4	9256.98	112400.66	2645.36	932.33	267.30	484.69	1228.63	142.32	620.57	133.02	37.37	150.70	21.92	129.29	28.15	76.68	12.18	69.29	12.58	291.21	355.73
SM42	CG4	5684.41	157873.86	3038.02	88.18	351.74	50.30	99.03	10.00	41.76	9.13	1.86	8.70	1.16	10.03	2.24	7.85	0.80	6.27	1.16	3.19	50.17
SM43	CG4	15079.09	124432.76	2734.16	854.52	245.63	435.16	847.52	102.84	444.82	103.13	25.78	115.76	16.56	106.21	21.88	72.26	9.06	65.69	10.97	133.04	548.54
SM44	CG4	6336.88	164399.53	3055.96	211.28	361.90	120.88	284.42	26.44	105.01	24.17	5.10	24.66	4.02	26.85	5.60	16.96	2.55	16.06	2.70	19.43	166.76
SM45	CG4	6483.23	151715.75	3375.95	488.77	362.57	267.08	605.57	69.53	291.01	70.61	15.48	72.18	10.33	66.16	13.44	41.48	5.77	36.68	5.67	166.78	235.89
SM46	CG4	7387.93	149814.56	3274.13	326.11	352.88	175.91	401.96	42.38	184.91	43.18	9.05	46.34	6.25	39.19	8.64	24.50	3.37	25.39	4.11	58.16	149.83
SM47	CG4	7172.67	148508.59	3150.97	363.13	301.44	199.70	443.04	49.47	209.15	43.06	11.51	51.80	7.28	47.60	10.04	30.99	4.09	30.21	5.01	80.08	330.05
SM48	CG4	4080.12	155801.97	3033.71	304.85	332.30	175.50	398.82	41.55	167.41	37.12	9.37	37.68	5.80	37.80	7.88	24.48	3.25	25.79	4.04	37.89	186.46
SM49	CG4	7128.31	165940.64	3113.70	107.84	336.77	56.83	114.48	11.40	45.63	9.51	2.19	10.67	1.76	10.39	2.57	8.29	1.16	10.23	1.43	5.18	122.49
SM50	CG4	4774.02	139318.50	3306.49	343.86	319.33	190.85	454.48	53.25	221.41	51.13	11.63	53.50	7.17	45.46	9.96	30.21	3.81	29.11	4.43	107.85	181.68
SM51	CG4	5516.80	144905.42	3477.20	202.13	356.31	114.00	260.21	30.44	129.29	28.50	6.77	29.18	4.59	28.96	5.99	16.79	2.14	15.23	2.24	57.6	105.87
SM52	CG4	6699.38	169414.97	2849.51	887.46	142.34	533.49	1233.78	116.72	469.44	102.23	21.23	114.55	16.47	108.87	23.34	67.58	10.17	69.94	12.53	25.23	161.13
SM53	CG4	6308.89	144225.11	2991.63	413.60	317.97	236.93	588.55	60.67	254.65	57.39	12.88	55.82	8.74	53.63	11.37	36.22	5.48	34.05	5.68	61.45	307.31
SM54	CG4	5355.96	129142.12	2966.31	310.94	307.70	169.52	406.97	44.43	189.92	44.14	11.09	45.36	6.36	39.32	9.19	26.23	3.54	27.04	4.15	84.38	260.42
SM55	CG4	6408.67	141155.20	3181.13	756.88	323.81	438.47	951.22	115.79	493.75	115.91	26.68	117.01	15.46	99.75	21.93	63.57	8.26	57.32	8.45	93.21	172.91
SM56	CG4	6100.74	118550.61	2776.30	407.90	285.50	227.16	537.08	57.44	251.28	60.41	11.95	61.35	9.20	50.97	10.33	37.61	4.81	31.10	5.10	131.45	398.67
SM57	CG4	6596.29	135664.27	3171.19	990.00	350.96	542.78	1213.53	142.23	584.09	129.30	30.78	141.62	19.49	131.62	25.70	79.20	11.13	70.31	11.79	180.77	396.84
SM58	CG4	5035.51	158237.59	3200.19	528.64	304.78	275.77	634.56	68.69	294.56	63.77	15.57	71.76	10.21	67.18	13.16	44.31	5.35	40.46	6.29	48.8	159.45
SM59	CG4	6620.49	149257.91	3395.90	412.56	496.45	252.49	610.50	59.43	245.89	50.84	12.19	54.68	8.07	48.61	10.47	34.13	3.90	30.78	4.66	31.67	139.07
SM60	CG4	7011.52	150115.94	3356.64	152.27	377.51	86.62	176.28	19.99	83.46	16.38	4.86	20.79	2.39	18.90	4.09	11.29	1.76	11.21	2.25	30.02	137.99
SM61	CG4	4835.32	140933.45	2978.71	299.67	322.45	161.74	363.74	40.08	174.13	39.31	11.27	40.66	5.79	36.84	8.01	26.18	3.81	24.05	3.64	71.51	356.61
SM62	CG4	7622.76	140992.61	3153.96	321.62	335.52	181.31	397.86	49.21	199.76	45.70	10.84	50.26	6.89	39.82	8.87	23.98	3.72	25.50	4.45	96.19	169.21
SM63	CG4	6054.48	157032.72	3592.59	61.08	342.73	32.31	71.84	7.77	30.84	8.02	1.73	7.35	1.07	8.23	1.59	4.35	0.71	4.55	0.78	15.57	58.78
SM64	CG4	2395.75	161085.13	3023.64	765.62	259.08	415.75	880.59	108.83	469.95	103.74	24.53	112.89	15.22	97.96	20.12	64.07	8.34	53.84	7.86	144.78	296.94
SM65	CG4	6575.48	143567.69	3311.40	119.66	384.18	59.36	104.96	12.32	55.57	10.66	2.45	12.31	1.83	13.66	2.91	9.61	1.30	9.18	1.59	11.21	99.18
SM66	CG4	10599.25	149652.59	3071.39	733.95	384.73	414.61	860.69	96.24	422.86	91.94	21.52	99.06	14.17	91.36	18.55	61.32	8.39	50.99	8.63	47.83	217.97
SM67	CG4	4004.48	173403.23	3135.33	80.11	331.62	42.03	84.02	7.35	27.36	5.92	1.17	7.41	1.28	9.14	1.65	6.10	0.85	7.35	1.29	4.01	97.26
SM68	CG4	4386.63	146359.98	3027.98	220.47	268.38	115.25	256.48	28.75	113.52	27.97	6.25	28.80	4.40	28.77	5.70	19.64	2.53	18.27	2.87	42.48	233.23
SM69	CG4	10851.89	139470.92	3227.32	592.91	331.71	305.03	693.64	79.07	334.47	75.58	17.72	82.70	12.20	73.88	15.90	49.02	6.53	46.91	7.05	88.55	180.27
SM70	CG4	6753.68	163162.41	3189.48	57.12	343.43	32.81	51.53	5.21	21.21	4.21	0.97	4.57	0.61	4.69	1.24	4.20	0.56	4.89	0.87	1.76	79.85
SM71	CG4	8341.81	136644.63	3303.91	528.50	328.89	308.93	485.04	56.31	244.60	48.54	11.71	60.80	8.19	53.19	11.94	37.24	5.11	33.31	5.55	72.1	150.5
SM72	CG4	15718.05	135531.88	3103.16	652.71	370.36	358.30	817.71	85.23	363.58	78.76	18.48	87.34	11.84	78.66	16.41	50.06	6.86	43.34	7.15	35.64	191.8
SM73	CG4	12645.08	136813.23	3070.69	709.85	356.01	391.13	870.83	103.62	437.89	97.62	23.84	99.07	14.71	95.61	18.14	58.92	7.00	48.83	7.55	115.93	147.66
SM74	CG4	10953.85	146189.59	2913.76	978.54	250.21	535.17	987.99	114.25	480.21	103.79	22.97	116.84	16.14	104.71	22.03	71.85	9.69	66.26	10.06	64.34	235.88
SM75	CG4	7780.63	144219.08	3215.64	182.96	365.49	96.85	235.17	23.77	101.81	22.43	5.51	25.46	3.63	22.5	4.53	15.69	2.08	15.09	2.64	43.19	195.45
SM76	CG4	4072.92	168579.56	3448.56	50.48	335.26	21.38	44.21	4.07	16.34	3.23	0.78	4.09	0.65	4.02	1.06	3.96	0.52	4.1	0.71	1.96	62.97
SM77	CG4	4282.77	157583.77	2858.97	120.87	290.03	66.39	151.71	15.62	65.25	14.54	3.13	15.18	2.34	15.77	3.14	10.28	1.3	10.07	1.6	27.24	147.17
SM78	CG4	5302.85	137858.64	2975.26	192.07	306.36	97.71	204.71	21.55	87.22	22.43	5.03	24.99	3.5	22.71	4.91	16.31	2.04	17.4	2.75	28.94	296.03
SM79	CG4	6707.68	152110.94	3067.36	208.01	339.39	114.85	231.09	24.33	98.89	21.65	4.91	23.73	3.31	22.27	5.15	16.95	2.06	18.41	2.77	20.11	161.8
SM80	CG4	7818.14	143047.13	2904.05	396.82	301.11	209.31	447	47.56	199.48	43.9	10.36	50.49	7.17	44.03	10.17	30.6	4.36	31.63	5.2	34.18	272.39
SM81	CG4	9068.37	152302.08	2934.45	298.63	314.25	160.09	364.71	36.7	146.77	35.83	7.63	38.46	5.69	35.33	7.7	25.15	3.32	25.51	4.35	38.84	230.1
SM82	CG4	9742.13	148350.73	2991.86	517.61	306.27	279.86	559.46	67.43	274.43	59.34	13.95	69.11	9.43	61.07	13.39	39.83	5.51	38.76	5.84	39.56	173.67
SM83	CG4	6433.59	158284.52	3576.57	67.91	348.87	37.33	81.13	9.8	40.29	9.42	2.13	9.22	1.36	8.25	1.78	5.65	0.66	5.82	0.72	25.66	67.57
SM84	CG4	7757.56	117983.34	2999.48	982.53	86.8	471.89	781.77	119	516.49	120.09	27.09	129.85	17.78	115.2	24.47	76.84	9.31	66.37	9.42	121.9	210.72
SM85	CG4	8053.72	161326.67	3179.52	218.54	304.4	114.89	249.33	24.04	99.65	21.56	4.94	24.42	3.78	24.49	5.24	17.67	2.46	18.13	3.02	6.74	126.87
SM86	CG4	3231.58	159673.23	3265.14	15.75	328.81	6.91	13.18	1.36	5.15	1.16	0.22	1.25	0.2	1.02	0.37	1.21	0.14	1.41	0.14	0.84	40.17
SM87	CG4	8419.17	163531.48	3500.73	98.52	348.71	53.01	113.45	10.98	46.51	9.76	2.37	10.7	1.55	10.68	2.37	7.26	1.11	7.89	1.53	7.34	65.67
SM88	CG4	77.73	149370.39	3526.75	53.35	550.19	26.18	51.47	6.97	26.7	8.65	1.46	6.83	1.16	6.72	1.3	3.65	0.53	2.96	0.53	13.32	66.48
SM89	CG4	8312.13	159247.94	3059.23	191.35	369.36	101.93	206.24	20.1	81.55	18.26	3.5	20.15	3.02	19.62	4.61	15.41	2.06	14.92	2.76	11.55	163.85
SM90	CG4	4038.56	167495.48	3456.2	43.68	303.22	20.34	38.03	4.02	14.84	3.26	0.82	3.89	0.47	4	0.9	3.35	0.44	3.32	0.6	1.95	49.42
SM91	CG4	6598.12	154314.8	3184.68	664.61	376.67	394.66	914.95	90.26	363.64	83.8	17.71	89.24	12.31	77.14	16.79	54.13	7.11	54.47	8.59	45.27	271.59
SM92	CG4	11295	152608.48	3312.18	557.3	406.98	329.74	665.1	76.6	317.51	70.4	15.56	74.51	11	68.77	14.24	43.5	6.31	42.89	6.96	67.72	225.38